Example 2.19 Use the method of undetermined coefficients to find the sum of $1+2+3+\cdots+(n-1)+n, n \in \mathbb{N}$

Solution:

$$\begin{aligned}
\text{Let } S(n) &= n+(n-1)+(n-2)+\cdots+2+1 \\
&= n+(n-1)+(n-2)+\cdots+[n-(n-2)]+[n-(n-1)] \\
&= n\left[1+\frac{n-1}{n}+\frac{n-2}{n}+\cdots+\frac{n-(n-2)}{n}+\frac{n-(n-1)}{n}\right] \\
&\leq n[1+1+\cdots+1] \quad \text{since } \frac{n-1}{n}<1,\ \frac{n-2}{n}<1,\cdots \\
\text{Thus, } S(n) &\leq n^2.
\end{aligned}$$

$$\begin{aligned}
\text{Let } S(n) &= a+bn+cn^2, \text{ where } a,b,c, \in \mathbb{R}. \\
\text{Now, } S(n+1)-S(n) &= n+1 \\
a+b(n+1)+c(n+1)^2-[a+bn+cn^2] &= n+1 \\
b+2cn+c &= n+1
\end{aligned}$$

Thus, $b+c=1$ and $2c=1$ (Equating like coefficients) which give $b=\frac{1}{2};\ c=\frac{1}{2}$

Now, $S(1)=1,\ \ a+b+c=1$ which gives $a=0$

Hence, $S(n)=\frac{1}{2}n+\frac{1}{2}n^2=\frac{n(n+1)}{2}, n \in \mathbb{N}$.

Example 2.20 Find the roots of the polynomial equation $(x-1)^3(x+1)^2(x+5)=0$ and state their multiplicity.

Solution:
Let $f(x)=(x-1)^3(x+1)^2(x+5)=0$. Clearly, we have $x=1,-1,-5$.
Hence, the roots are 1 with multiplicity 3, -1 with multiplicity 2 and -5 with multiplicity 1.



When the root has multiplicity 1, it is called a simple root.

Example 2.21 Solve $x=\sqrt{x+20}$ for $x \in \mathrm{R}$.

Solution:
By the definition of square root, $\sqrt{x+20} \geq 0$.
Since $x=\sqrt{x+20}$, x is positive.
Now squaring we get $x^2=x+20$. $x^2-x-20=0$
$(x-5)(x+4)=0$, which gives $x=5, x=-4$
Since, x is positive, the required solution is $x=5$.